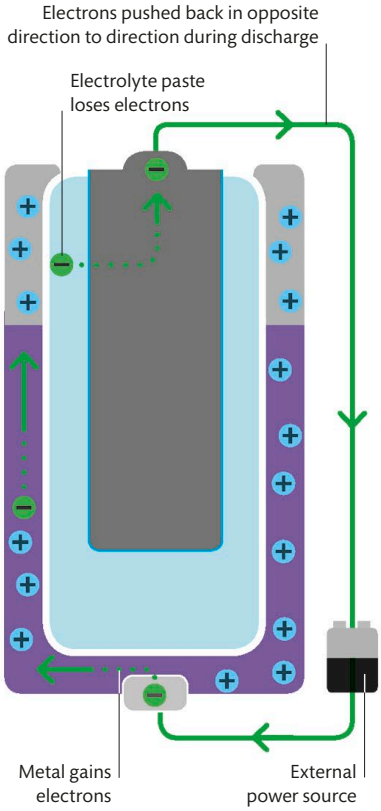
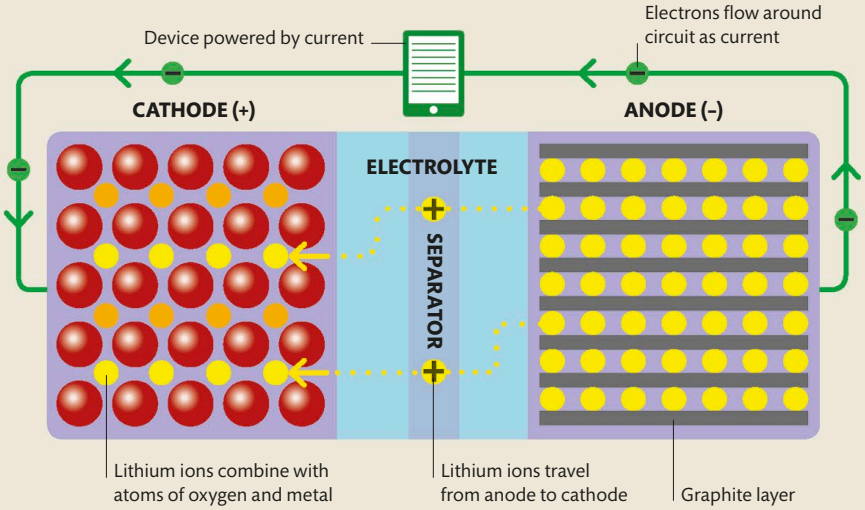
How a battery recharges
When a battery is plugged into a charger, a current passes through it in the opposite direction to that produced when the battery was discharging. This moves electrons back to where they started, recharging the battery.



Lithium-ion batteries

Found in smartphones and many other devices and machines, including electric cars, lithium-ion (Li-ion) batteries use the large amounts of energy present in the highly reactive metal lithium. Lithium's low weight but high energy density produces a good power-to-weight ratio for batteries that can withstand hundreds of discharge and recharge cycles.

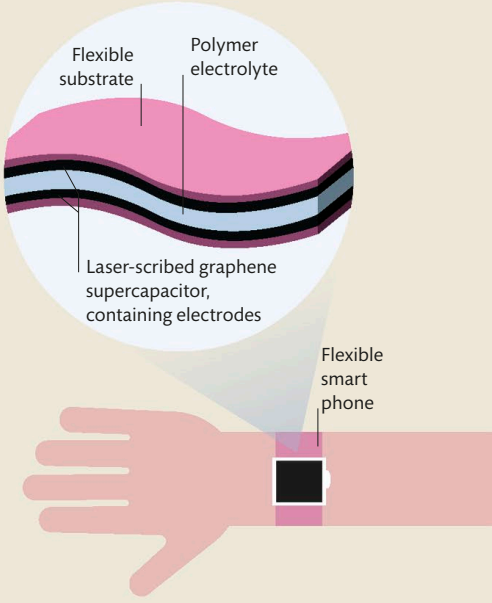


How a lithium-ion battery works
Lithium ions flow through the electrolyte to the cathode during discharge while the free electrons flow through the external circuit, providing power. Recharging returns the lithium ions and electrons.



Batteries of the future

A great deal of research is focused on battery development. One innovation that may lead to faster-charging, longer-lasting batteries uses a solid state alkali metal rather than the liquid or gel electrolytes in Li-ion cells. Flexible batteries using devices called supercapacitors that can be recharged in seconds might revolutionize wearable and portable technology.



Supercapacitor
Electric charge is stored as a coating of ions on a supercapacitor's electrode layers, which are separated by an electrolyte made of a flexible polymer.

WHAT IS THE WORLD'S LARGEST BATTERY?

Tesla's gigantic lithium-ion battery in South Australia covers 2.5 acres (1 hectare) and provides 129 MWh (see p.10) of electricity.